Xinyu Tian

LinkedIn | GitHub | Website

SUMMARY

M.Sc. candidate at NUS with computer vision research experience at A*STAR IME in SEM defect detection. Research experience in RGB-D salient object detection. Proficient in Python, PyTorch, OpenCV.

EDUCATION

National University of Singapore (NUS), Singapore ***08/2025-01/2027(Expected)***

Master of Science in Physics for Technology

Relevant Courses: Machine Learning, Complex Systems Analysis, Quantum Computation, AI for Optics.

Focus Areas: Computer Vision, Machine Learning.

Southwest Jiaotong University, Chengdu, China ***09/2021-06/2025***

Bachelor of Science in Applied Physics **GPA: 86.85/100(Top15%)**

Relevant Courses: Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Intelligent Detection and Control, Basics of Computer Programming, Digital Image Processing, Signals and Systems.

A+ grades in multiple programming courses, including Basics of Computer Programming and Computational Physics.

EXPERIENCE

A*STAR - Agency for Science, Technology and Research IME | *Computer Vision Research Intern* *08/2025-04/2026*

SEM-Based Wafer Defect Detection System

- Developed an **end-to-end CNN-based defect detection pipeline** to identify AOGs from wafer SEM surface images.
- Collected and cleaned **288 high-resolution SEM images** through manual acquisition on SEM equipment, and **annotated pixel-level masks** in CVAT to build a labeled dataset.
- Benchmarked a classical intensity-based baseline, plain UNet, and a ResNet34-based UNet under identical settings; the best model achieved test **IoU 0.692, Dice 0.814, and Recall 0.852**, **a 39.5% relative IoU improvement** over plain UNet.
- Deployed the model as an application prototype that automatically reports **defect count, location, and area**, achieving **over 85% accuracy** against manual ground truth, and generates **batch-level statistical charts** for faster, more objective materials characterization.

SELECTED PROJECTS

Dual-Domain Personal Website | *Independent Developer* *04/2026-07/2026*

- **Independently designed, built and deployed** a bilingual (EN/ZH) personal website using **Next.js, TypeScript, and Tailwind CSS**, serving two domains from one codebase with domain-specific content modes and metadata.
- Built **15+ reusable components** and structured content; integrated a **YouTube API feed** with caching and error handling, and prototyped an **interactive MapLibre travel map** linked to photo albums.
- Worked in an **AI-assisted workflow** with **Claude Code and Codex**, using written project guidance, **Git version control**, and lint checks throughout iterative development.

RGB-D Salient Object Detection via Multi-Modal Feature Fusion | *Graduation Thesis* *12/2024-06/2025*

Southwest Jiaotong University

- Reproduced and optimized CMiNet for **RGB-D salient object detection**, training on NJUD+NLPR and COME15K.
- Evaluated performance on 5 benchmark test sets using **MAE, F-measure, and PR curves**; ablation studies showed ≥2.7% F-measure improvement from critical modules.
- Compared with 12 fusion-based baselines on identical test sets, **achieving stronger results on four core metrics**, including an 8.5% MAE reduction versus BBS-Net.

YOLOv5-Based Surface Defect Detection for Industrial Inspection | *Team Leader* *10/2024-12/2024*

Project-oriented AI Course, Southwest Jiaotong University

- Implemented and trained a **YOLOv5-based defect detection pipeline** in Python and PyTorch using the NEU-DET dataset, which contains 1,800 grayscale images across 6 defect classes, for surface defect inspection.
- Prepared dataset splits and evaluated performance using mAP, F1-score, precision, recall, achieving mAP 0.772, and maximum recall 0.97, demonstrating the feasibility of **real-time defect detection** for industrial inspection tasks.

SKILLS

Programming: Python, C/C++, TypeScript, MATLAB

CV/ML: PyTorch, OpenCV, TensorFlow, NumPy

Tools: Git, Jupyter, Linux, Docker, Vercel, Next.js

AI Tools: Codex, Claude Code, Copilot, Cursor